

GPT and BERT

DATA 443/643 · Week 3, note 3.4

Week 3

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Ideas from Princeton COS 484 L10 and CS224N pretraining (GPT vs BERT). Original slides; those courses are not copied.

Lecture plan

- 1 Static table versus contextual vector

- 2 Causal next-token versus masked tokens

- 3 Same width, different mask and loss

- 4 What a project is allowed to stream

First time: what / why / formula

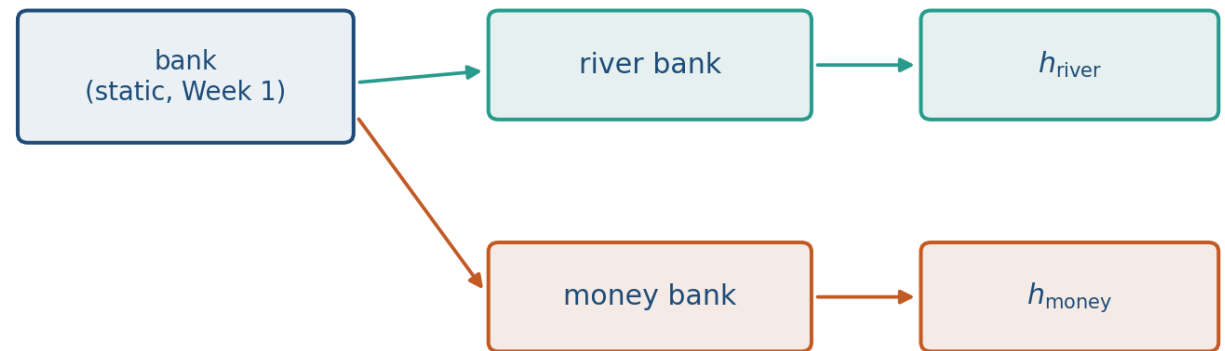
- What / why. GPT: causal next-token (left to right). BERT: masked tokens, bidirectional context. Both produce contextual vectors, not a static table (Week 1). Language modeling (GPT) is the pretrain that later becomes ChatGPT. BERT-style MLM is a bidirectional encoder for classification and span tasks.
- Architecture. Same transformer block. GPT masks the future. BERT masks random token ids and reconstructs them. Heads differ (LM vs CLS/span).
- Formula. GPT: $\prod_t p(w_t \mid w_{<t})$. BERT: reconstruct masked positions, not a full joint factorization of the sentence.
- Tradeoffs. + Contextual geometry. – GPT cannot see the future inside a prompt token; BERT is not a natural generator. “Transformer” is the block; GPT/BERT is the objective.

How. Train on unlabeled text. After pretrain, GPT generates; BERT is usually fine-tuned with a small head.

The same type, two vectors

- Week 1: bank is one row.
- A transformer: river bank is not money bank.
- Pretrain once, fine-tune many times.

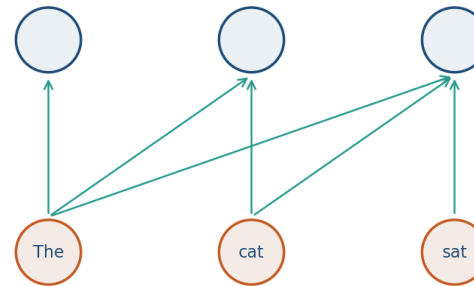
A transformer embedding is contextual: the same type, two vectors



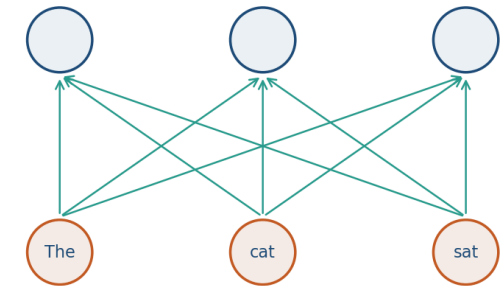
Same block, different mask

- GPT: look left, predict the next token.
- BERT: look both ways, fill [MASK].
- Those are two different uses of the word mask.

GPT: causal (next token)



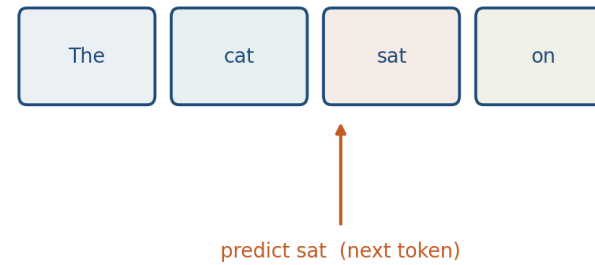
BERT: bidirectional (masked token)



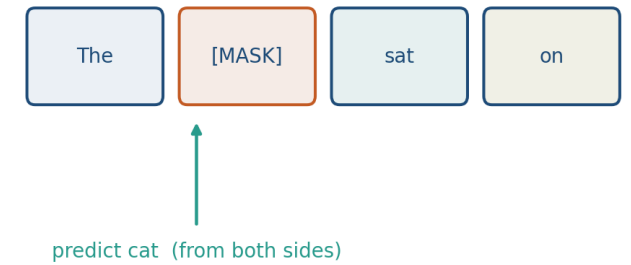
Two objectives

- GPT: $p(x_t \mid x_{1:t-1})$.
- BERT: $p(\text{masked token} \mid \text{the rest})$.
- Do not fine-tune BERT as if it were GPT.

GPT: causal language model



BERT: masked language model



What you can ship

Decoder (GPT-style)

Generate, chat, RAG, tools. The default stack for this course.

Encoder (BERT-style)

Classify, tag, embed a span. Cannot stream a paragraph.

A causal 2 by 2

$$S_{\text{GPT}} = \begin{bmatrix} 1 & -\infty \\ 3 & 4 \end{bmatrix}$$

Row 1 softmax is [1, 0]. Token 1 cannot see token 2.

Row 2 is softmax([3, 4]) about [0.27, 0.73].

BERT softmaxes the unmasked S. Token 1 does look ahead.

What to take from this lecture

Context

Embeddings are now a function of the sentence.

Fork

Causal LM versus MLM. Same d , different job.

Project

If it must stream tokens, it is not BERT.